

ODM 3D Mapping (Digital Twin)

Goal:

Create a digital twin dashboard that can process aerial images into 3D models using OpenDroneMap and display it in a web interface

Code Structure:

- Backend (Open Drone Map integration)
- Frontend (CesiumJS? Three.js? Need to research this more)

Features:

- Dashboard that can display the 3D model created from ariel drone data
 - o User should be able to navigate in the model from the frontend

Long-Term Features:

- Take other data from All Seeing Research and import it into our system so the farmer can see data about their entire farm from a simple application.

Resources:

- OpenDroneMap Docs <https://www.opendronemap.org/>
- CesiumJS Docs <https://cesium.com/learn/cesiumjs-learn/>
 - o We have tried using the cesium plugin for unreal before, but it was not used much, documentation on it can be found here:
<https://missourimrr.github.io/docs/simulation/cesiumforunreal/>
- Three.js Docs <https://threejs.org/docs/>
- Datasets:
 - o https://drive.google.com/drive/folders/1JKjYn_IPnzhHSg_5K3Ou3kcuac4gvPSI
 - o <https://mailmissouri.sharepoint.com/sites/MulticopterDesignTeam-Grp/Administration/Forms/AllItems.aspx?id=%2Fsites%2FMulticopterDesignTeam%2DGrp%2FAdministration%2FMedia%2FFlight%20Days%2F26%2D27%2F2026%2D08%2D29%2DSparrow%2DTesting&viewid=934cefe8%2D1c60%2D452c%2Dada6%2D7842fb155ddc&d=w7d7e48b9a3894375ab5720dda1fcd328&csf=1&CID=65f150c0%2D88bb%2D455f%2D9db5%2D21166ee72148&FolderCTID=0x012000FFCA1ABA01F90B4F8BA525AC1F600A6B>

Timeline:

- Make sure everyone has access to the GitHub/SharePoint
- Research ODM and frontend 3D libraries (CesiumJS/Three.js) to find a way to implement basic 3D mapping.
- Make sure this all can run on Ricky (Bay Computer) or team members computers.

- Use our ODLIC (mapping) dataset to test the 3D mapping or research what data we would need.
 - Note: All Seeing Research is supposedly sending us data soon.
- Write the software stack and have it approved by Zach and Matthew
- Begin development!

***NOTE:** These task lists are just suggestions for who does what, do not feel obligated to strictly follow them. Just let your lead know what you want to do. You will *never* have to do anything you are not interested in.

Timelines*		
Task List 1 (OpenDroneMap)	Task List 2 (3D Viewer)	Task List 3 (Dashboard)
Research OpenDroneMap, Threes.js, and CesiumJS	Research OpenDroneMap, Threes.js, and CesiumJS	Research OpenDroneMap, Threes.js, and CesiumJS
Get stack approved by Zach/Matthew	Get stack approved by Zach/Matthew	Get stack approved by Zach/Matthew
Get ODM running on sample images	Get a basic viewer with a sample model and basic navigation	Create a simple dashboard skeleton
Run ODM on real data	Add zoom/pan/click controls, stress test	Build farm selection logic and other requirements (TBD)
Export ODM to the 3D viewer's input	Any debugging/performance issues that arise from the real data	TBD based on All Seeing Research's new requirements
Turn entire functionality into a script	Make a dedicated scaling and camera speeds	TBD